

Solutions – Practice Problem Set 1

ECON 311 — Intermediate Macroeconomics — Fall 2026

Problem 1 Growth-rate arithmetic.

Background: The key rule is that if $z = u^a v^b$, then $g_z = a g_u + b g_v$, where growth rates of products add and growth rates of ratios subtract.

Solution

(a) $z = x^2 y^3$

$$\Rightarrow g_z = 2g_x + 3g_y$$

(b) $z = \frac{x^{0.6}}{y^{0.4}} = x^{0.6} y^{-0.4}$

$$\Rightarrow g_z = 0.6 g_x - 0.4 g_y$$

(c) $z = x^{-0.5} \left(\frac{1}{w}\right)^{0.3} y^{0.8} = x^{-0.5} w^{-0.3} y^{0.8}$

$$\Rightarrow g_z = -0.5 g_x - 0.3 g_w + 0.8 g_y$$

(d) $z = \left(\frac{xy}{w^2}\right)^{1/2} = x^{1/2} y^{1/2} w^{-1}$

$$\Rightarrow g_z = \frac{1}{2}g_x + \frac{1}{2}g_y - g_w$$

Problem 2 Returns to scale.

Method: Scale both inputs by $\gamma > 0$: replace $K \rightarrow \gamma K$ and $L \rightarrow \gamma L$. If $F(\gamma K, \gamma L) = \gamma^\lambda F(K, L)$, the function has: $\lambda > 1$ (IRS), $\lambda = 1$ (CRS), $\lambda < 1$ (DRS). For a Cobb–Douglas $Y = AK^\alpha L^\beta$: check $\alpha + \beta$.

Solution

(a) $Y = \bar{A}K^{1/4}L^{3/4}$: exponents sum to $\frac{1}{4} + \frac{3}{4} = 1$. **Constant returns to scale (CRS).**

(b) $Y = K^{0.6}L^{0.6}$: exponents sum to $0.6 + 0.6 = 1.2 > 1$. **Increasing returns to scale (IRS).**

(c) $Y = K^{1/3}L^{2/3} + \bar{A}$: the exponents of the Cobb–Douglas part sum to 1, but there is an additive constant, so we cannot read off RTS from the exponents alone. Apply

the definition:

$$F(\gamma K, \gamma L) = (\gamma K)^{1/3}(\gamma L)^{2/3} + \bar{A} = \gamma K^{1/3}L^{2/3} + \bar{A}.$$

Compare with $\gamma F(K, L) = \gamma K^{1/3}L^{2/3} + \gamma \bar{A}$. The difference is

$$F(\gamma K, \gamma L) - \gamma F(K, L) = \bar{A} - \gamma \bar{A} = (1 - \gamma)\bar{A}.$$

For $\gamma > 1$, this is negative, meaning $F(\gamma K, \gamma L) < \gamma F(K, L)$. **Decreasing returns to scale (DRS).**

Economic interpretation: The constant $\bar{A}$ (e.g., a fixed factor such as land or a licence) is not scaled, so doubling inputs produces less than double the output.

(d) $Y = K + K^{0.5}L^{0.5}$: apply the definition.

$$F(\gamma K, \gamma L) = \gamma K + (\gamma K)^{0.5}(\gamma L)^{0.5} = \gamma K + \gamma K^{0.5}L^{0.5} = \gamma(K + K^{0.5}L^{0.5}) = \gamma F(K, L).$$

Constant returns to scale (CRS).

Problem 3 National income accounting.

Setup summary:

- Lumber company: pays \$18,000 wages; sells 400 m³ to furniture mfr. at \$60/m³ = \$24,000; sells planks to households for \$9,000.
- Furniture manufacturer: buys timber \$24,000; pays wages \$30,000; makes 50 chairs. Sales: 35 to households × \$1,200 = \$42,000; 10 to government × \$1,200 = \$12,000; 5 exported × \$1,400 = \$7,000. Total revenue = \$61,000.

Solution

(a) Expenditure approach

Count only *final* goods (those not used as intermediate inputs):

$$C = \text{planks to households} + \text{chairs to households} = 9,000 + 42,000 = \$51,000$$

$$G = \text{chairs to government} = \$12,000$$

$$NX = \text{exported chairs} = \$7,000$$

$$I = 0 \quad (\text{no capital investment described})$$

$$GDP = C + G + NX = 51,000 + 12,000 + 7,000 = \$70,000$$

(b) Income approach

Labour income:

$$18,000 + 30,000 = \$48,000$$

Capital income (profit) — each firm:

Lumber company revenue = 24,000 + 9,000 = \$33,000. No intermediate inputs. Profit = 33,000 – 18,000 = \$15,000.

Furniture manufacturer revenue = 42,000 + 12,000 + 7,000 = \$61,000. Costs: labour \$30,000 + timber \$24,000 = \$54,000. Profit = 61,000 – 54,000 = \$7,000.

$$GDP = 48,000 + 15,000 + 7,000 = \$70,000$$

(c) Value-added approach

Value added equals revenue minus the cost of intermediate inputs.

Firm	Revenue	Intermediate inputs
Lumber company	\$33,000	\$0
Furniture manufacturer	\$61,000	\$24,000
Value added: lumber = \$33,000; furniture = \$37,000		

$$GDP = 33,000 + 37,000 = \$70,000$$

(d) All three methods yield \$70,000. ✓

Problem 4 Real GDP, nominal GDP, and the GDP deflator.**Solution****(a) Nominal GDP in Year 2**

Final goods only, at Year-2 prices:

$$C = \text{planks to households} + \text{chairs to households} = 11,500 + 35 \times 1,500 = 11,500 + 52,500 = \$64,000$$

$$G = 10 \times 1,500 = \$15,000$$

$$NX = 5 \times 1,700 = \$8,500$$

$$\text{Nominal GDP}_2 = 64,000 + 15,000 + 8,500 = \boxed{\$87,500}$$

(b) Real GDP in Year 2 (Year-1 base prices)

Quantities are identical to Year 1; prices are Year-1 prices:

$$\text{Real GDP}_2 = \text{GDP}_1 = \$70,000$$

$$\text{Real GDP growth rate} = \frac{70,000 - 70,000}{70,000} = \mathbf{0\%}.$$

(c) GDP deflator for Year 2

$$\text{GDP Deflator}_2 = \frac{\text{Nominal GDP}_2}{\text{Real GDP}_2} = \frac{87,500}{70,000} \approx 1.250$$

$$\text{Percentage change in the price level} = (1.250 - 1.000) \times 100\% = \mathbf{25\%}.$$

(d) Economic interpretation

Real GDP measures the value of output at constant (base-year) prices, so it captures only changes in the *quantity* of goods produced. Since all quantities remained the same in Year 2, real GDP did not change. Nominal GDP rose solely because prices rose. The GDP deflator isolates this pure price change: the general price level rose by 25% between Year 1 and Year 2.

Problem 5 Economic convergence.**Solution****(a) Catch-up formula**

Set $Y_t^A = Y_t^B$:

$$5,000 (1.04)^{t^*} = 24,000 (1.02)^{t^*}$$

$$\left(\frac{1.04}{1.02}\right)^{t^*} = \frac{24,000}{5,000} = 4.8$$

$$t^* \cdot \ln\left(\frac{1.04}{1.02}\right) = \ln(4.8) \Rightarrow t^* = \frac{\ln(4.8)}{\ln(1.04/1.02)} = \frac{1.5686}{0.019608} \approx \mathbf{80.8} \text{ years}$$

Country Alder catches up approximately in the year $2000 + 81 = \mathbf{2081}$.

(b) Accelerated early growth

For the first 15 years Alder grows at 7%; thereafter at 4%. Country Birch grows at 2% throughout. Set them equal at time T (years after 2000, where $T > 15$):

$$5,000 (1.07)^{15} (1.04)^{T-15} = 24,000 (1.02)^T$$

$$\frac{(1.04)^T}{(1.04)^{15} (1.02)^T} = \frac{24,000}{5,000 (1.07)^{15}}$$

$$\left(\frac{1.04}{1.02}\right)^T = \frac{24,000}{5,000 (1.07)^{15}} (1.04)^{15}$$

Calculate the numerical constants: $(1.07)^{15} \approx 2.7590$, $(1.04)^{15} \approx 1.8009$.

$$\left(\frac{1.04}{1.02}\right)^T = \frac{24,000}{5,000 \times 2.7590} \times 1.8009 = \frac{24,000 \times 1.8009}{13,795} \approx \frac{43,222}{13,795} \approx 3.1330$$

$$T = \frac{\ln(3.1332)}{\ln(1.04/1.02)} = \frac{1.1422}{0.019608} \approx \mathbf{58.3} \text{ years}$$

With the 7% early growth, Alder catches up approximately in year $2000 + 58 = \mathbf{2058}$ — about 23 years sooner.

(c) Economic interpretation

The accelerated early growth compresses the initial income gap much faster, so Alder needs fewer years at the moderate 4% rate to close the remaining gap. In practice, a country could achieve 7% growth through institutional reforms, technology transfer, foreign direct investment, or capital accumulation from a very low capital base (consistent with the Solow model: a poor country far below its steady state has a high marginal product of

capital and thus high growth rates).

Problem 6 From aggregate to per-capita variables.

Solution

(a) Divide both sides of $Y_t = AK_t^\alpha L_t^{1-\alpha}$ by L_t :

$$\frac{Y_t}{L_t} = A \frac{K_t^\alpha L_t^{1-\alpha}}{L_t} = A \frac{K_t^\alpha}{L_t^\alpha} = A \left(\frac{K_t}{L_t} \right)^\alpha = Ak_t^\alpha.$$

Therefore $y_t = Ak_t^\alpha$. □

(b) Start from the capital accumulation equation and substitute $I_t = \bar{s}Y_t$:

$$K_{t+1} = (1 - \delta)K_t + \bar{s}Y_t.$$

Subtract K_t from both sides:

$$\Delta K_{t+1} = \bar{s}Y_t - \delta K_t.$$

Divide both sides by L_t :

$$\frac{\Delta K_{t+1}}{L_t} = \bar{s} \frac{Y_t}{L_t} - \delta \frac{K_t}{L_t} = \bar{s} y_t - \delta k_t. \quad (*)$$

Now use the approximation for growth rates. Since $k_t = K_t/L_t$:

$$g_k \approx g_K - g_L = \frac{\Delta K_{t+1}}{K_t} - \bar{n}.$$

Multiplying both sides by k_t :

$$\frac{\Delta k_{t+1}}{1} = k_t \cdot g_k = \frac{\Delta K_{t+1}}{K_t} \cdot k_t - \bar{n} k_t = \frac{\Delta K_{t+1}}{L_t} - \bar{n} k_t. \quad (**)$$

Substituting (*) into (**):

$$\Delta k_{t+1} = \bar{s} y_t - \delta k_t - \bar{n} k_t = \bar{s} y_t - (\delta + \bar{n}) k_t. \quad \square$$

(c) Economic interpretation

- $\bar{s} y_t$: **actual investment per person.** This is the amount of new capital being added to the economy per worker each period.

- δk_t : **depreciation**. A fraction δ of the existing capital stock per worker wears out each period.
- $\bar{n} k_t$: **capital dilution**. Even if no capital depreciates, a growing population spreads the existing capital across more workers, reducing capital per person.
- The sum $(\delta + \bar{n})k_t$ is the **break-even investment**: the investment needed just to keep k constant.
- $\Delta k_{t+1} > 0$ when actual investment exceeds break-even investment; capital per worker rises. At steady state, $\bar{s} y^* = (\delta + \bar{n})k^*$.

Problem 7 Solow model: steady state.

Solution

(a) **Steady-state derivation** ($\alpha = 1/3$)

At steady state $\Delta k_{t+1} = 0$:

$$\bar{s} y^* = (\delta + \bar{n}) k^*.$$

Substitute $y^* = A k^{*1/3}$:

$$\bar{s} A k^{*1/3} = (\delta + \bar{n}) k^*.$$

Divide both sides by $k^{*1/3}$:

$$\bar{s} A = (\delta + \bar{n}) k^{*2/3} \quad \Rightarrow \quad k^{*2/3} = \frac{\bar{s} A}{\delta + \bar{n}} \quad \Rightarrow \quad k^* = \left(\frac{\bar{s} A}{\delta + \bar{n}} \right)^{3/2}$$

Then:

$$y^* = A k^{*1/3} = A \left(\frac{\bar{s} A}{\delta + \bar{n}} \right)^{1/2} = A^{3/2} \left(\frac{\bar{s}}{\delta + \bar{n}} \right)^{1/2}$$

(b) **Effect of an increase in $\bar{s}$**

Both k^* and y^* are increasing functions of $\bar{s}$ (the exponents are positive). A higher savings rate means more investment per worker each period, so the break-even condition is satisfied at a higher level of capital and output per worker.

Economic interpretation: When households save more, a larger share of output is channelled into investment. This raises the capital stock per worker and, through the production function, raises output per worker in the long run.

(c) **Effect of an increase in $\bar{n}$**

$(\delta + \bar{n})$ appears in the denominator (with a positive exponent), so both k^* and y^* *decrease* when $\bar{n}$ rises.

Economic interpretation: Faster population growth means each unit of new investment must be spread over more workers just to maintain capital per worker (the dilution effect is larger). This leaves less net investment for deepening capital, pushing the steady-state capital-labour ratio down.

Problem 8 Solow model: numerical dynamics.

Solution

(a) Steady-state values

Parameters: $A = 1$, $\bar{s} = 0.25$, $\delta = 0.06$, $\bar{n} = 0.02$, so $\delta + \bar{n} = 0.08$.

$$k^* = \left(\frac{0.25 \times 1}{0.08} \right)^{3/2} = (3.125)^{3/2} = 3.125 \times \sqrt{3.125} \approx 3.125 \times 1.7678 \approx \mathbf{5.523}$$

$$y^* = 1^{3/2} \left(\frac{0.25}{0.08} \right)^{1/2} = (3.125)^{1/2} \approx \mathbf{1.768}$$

(b) Capital dynamics starting from $k_1 = 6$

The accumulation equation is: $\Delta k_{t+1} = 0.25 k_t^{1/3} - 0.08 k_t$.

Period 1 $\rightarrow$ 2:

$$y_1 = 6^{1/3} \approx 1.8171, \quad \Delta k_2 = 0.25(1.8171) - 0.08(6) = 0.4543 - 0.4800 = -0.0257$$

$$k_2 = 6 - 0.0257 = 5.9743.$$

Period 2 $\rightarrow$ 3:

$$y_2 = 5.9743^{1/3} \approx 1.8149, \quad \Delta k_3 = 0.25(1.8149) - 0.08(5.9743) = 0.4537 - 0.4779 = -0.0242$$

$$k_3 \approx 5.9501.$$

Period 3 $\rightarrow$ 4: $k_3 \approx 5.9501$

$$y_3 \approx 5.9501^{1/3} \approx 1.8128, \quad \Delta k_4 \approx 0.25(1.8128) - 0.08(5.9501) = 0.4532 - 0.4760 = -0.0228$$

Period 4 $\rightarrow$ 5: $k_4 \approx 5.9273$

$$y_4 \approx 5.9273^{1/3} \approx 1.8108, \quad \Delta k_5 \approx 0.25(1.8108) - 0.08(5.9273) = 0.4527 - 0.4742 = -0.0215$$

Summary:

$$\Delta k_2 \approx -0.026, \quad \Delta k_3 \approx -0.024, \quad \Delta k_4 \approx -0.023, \quad \Delta k_5 \approx -0.022$$

(c) Pattern and explanation

The changes Δk_t are *negative* but *decreasing in absolute value*—capital per worker is falling, but more and more slowly. This occurs because $k_1 = 6 > k^* \approx 5.52$: the economy starts *above* its steady state. The break-even investment exceeds actual investment, so k falls each period. As k approaches k^* , the gap between actual and break-even investment shrinks, slowing the rate of decline.

(d) Above or below steady state?

$k_1 = 6 > k^* \approx 5.52$, so the economy is **above** its steady state. This is confirmed by $\Delta k_2 < 0$: capital per worker is declining toward k^* .

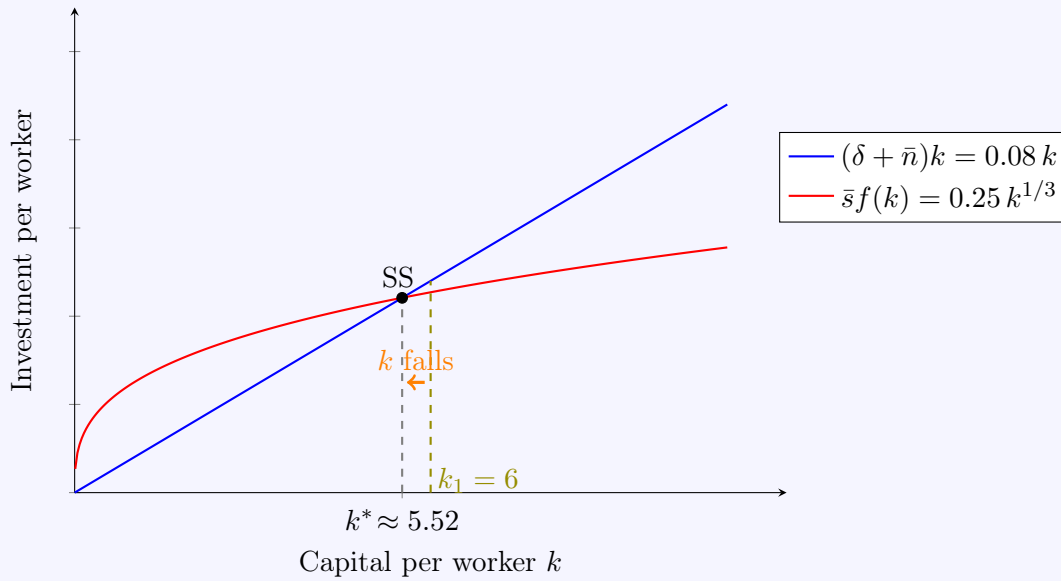
Problem 9 Solow diagram.

Solution

The two curves in the Solow diagram are:

- **Actual investment per worker:** $\bar{s}f(k) = 0.25k^{1/3}$ (concave, passes through the origin).
- **Break-even investment:** $(\delta + \bar{n})k = 0.08k$ (linear through the origin, slope = 0.08).

The steady state k^* is where the two curves intersect: $0.25k^{*1/3} = 0.08k^*$, giving $k^* \approx 5.52$.



(a) The concave curve is actual investment $\bar{s}f(k)$; the straight line is break-even investment $(\delta + \bar{n})k$. They cross at $k^* \approx 5.52$ (labelled “SS”).

(b) The initial position $k_1 = 6$ is marked to the right of k^* .

(c) Since $k_1 > k^*$, break-even investment exceeds actual investment at k_1 . The orange arrow shows k moving left (decreasing) toward k^* .

(d) **Effect of an increase in δ**

A higher depreciation rate raises the slope of the break-even investment line $(\delta + \bar{n})k$. The actual investment curve $\bar{s}f(k)$ does not change. The two curves now intersect at a lower value of k , so k^* falls. Economically, more capital is lost to depreciation each period, reducing the steady-state capital per worker.

Problem 10 Comparative statics and policy analysis.

Solution

(a) **Transitional dynamics after a savings-rate increase**

At the old steady state k^* , actual investment equals break-even investment: $\bar{s}y^* = (\delta + \bar{n})k^*$. When $\bar{s}$ rises to $\bar{s}'$, actual investment immediately becomes $\bar{s}'y^* > (\delta + \bar{n})k^*$, so $\Delta k_{t+1} > 0$: capital per worker begins to rise.

As k rises, $y = Ak^\alpha$ also rises (but at a decreasing marginal rate because $\alpha < 1$), while break-even investment $(\delta + \bar{n})k$ rises linearly. The gap between actual and break-even investment therefore narrows, and the growth of k slows down. The process continues until a new, higher steady state $k^{*'}$ is reached, at which point $\bar{s}'y^{*'} = (\delta + \bar{n})k^{*'}$ and

$\Delta k_{t+1} = 0$ again.

(b) Algebraic proof that higher $\bar{s}$ implies higher y^*

From Problem 7(a):

$$y^* = A^{3/2} \left(\frac{\bar{s}}{\delta + \bar{n}} \right)^{1/2}.$$

Taking the partial derivative with respect to $\bar{s}$:

$$\frac{\partial y^*}{\partial \bar{s}} = A^{3/2} \cdot \frac{1}{2} \left(\frac{\bar{s}}{\delta + \bar{n}} \right)^{-1/2} \cdot \frac{1}{\delta + \bar{n}} > 0.$$

Since A , $\bar{s}$, δ , $\bar{n}$ are all positive, this derivative is strictly positive. Therefore a higher $\bar{s}$ implies a higher y^* , all else equal. $\square$

(c) Conditional vs. absolute convergence

Absolute convergence would predict that all countries grow faster the poorer they are, unconditionally, and that all countries converge to the same income level in the long run. The Solow model predicts *conditional convergence*: a country converges to its own steady state y^* . Because y^* depends on country-specific parameters $(\bar{s}, \delta, \bar{n}, A)$, two countries with the same *current* income may be going in very different directions. A poor country converges to a high y^* if it has a high savings rate; a rich country with a low savings rate might converge to a lower y^* .

Conditional convergence is consistent with the empirical finding that countries converge when one controls for savings rates, population growth, and other structural parameters—but not unconditionally.

(d) Long-run growth and the need for technological progress

At the steady state of the basic Solow model, $\Delta k_{t+1} = 0$ and therefore $\Delta y_{t+1} = 0$. Output per capita is constant in the long run; the growth rate of output per capita is zero.

This is in tension with the experience of developed economies, which have sustained positive per-capita growth for over a century. The basic model implies that, without technological progress, capital accumulation alone cannot sustain long-run growth (diminishing returns to capital ensure that the marginal product eventually falls to the break-even level).

Extension: The standard resolution is to introduce **labour-augmenting technological progress** at rate $\bar{g}$: $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$ with $A_t = A_0(1 + \bar{g})^t$. The model then has a balanced growth path along which output per *effective* worker is constant, but output per *actual* worker grows at rate $\bar{g}$ —matching the long-run experience of developed economies.